Part 12 — The profession: publishing, writing, planning, the proposal and the application (weeks 62 to 66)

These five weeks are not about metabolic engineering. They are about how the work becomes a career: how a manuscript becomes a paper and who decides what at each stage, how to write so an argument survives a reader, how to plan a multi-year project when half the duration is waiting, how to assemble the capstone proposal, and how to apply. When part 6 went out I said these were the weeks a written module might add the most to, because nobody writes them down for an engineer arriving from another discipline — and that is still my view. This is the last part.

Two things carry over from the rest of the module and make these weeks different from generic career advice. The first is that **you already have the planning skill**: critical paths, buffers, lead times and risk registers are engineering practice, and a research proposal with a realistic timeline reads differently from one with only science. Week 64 makes that quantitative, and it is the one week here with real arithmetic in it — a schedule simulation I ran, whose central result is that a plan built from most-likely durations had a **1% chance** of being met.

The second is that the whole curriculum has been producing **artifacts**: notebooks, a reproduced design, flux maps, a host table, a paper log, a proposal. Weeks 65 and 66 are about pointing at them. The test the tracker keeps returning to is whether a sentence would survive unchanged in someone else's application; if it would, it is about ambition rather than evidence, and the artifacts are the cure.

Ch.	Week	Chapter
62	62	How publishing works: review, authorship and the record
63	63	Writing and talking about your work
64	64	Planning a multi-year project: the critical path and what you cannot control
65	65	The capstone proposal, section by section
66	66	Applications: labs, statement of purpose, interviews

How publishing works: review, authorship and the record

A PhD is judged largely by papers, so knowing how they are made is not administrative trivia — it is what lets you ask a prospective supervisor the questions that matter.

From manuscript to paper, and who decides what

Stage	Who decides	What is decided	Typical time
Submission	You	Venue, framing, which claims are made	—
Editorial (desk) check	Handling editor	Whether it is sent out for review at all	Days to two weeks
Reviewer selection and reports	Editor chooses; reviewers write	Whether the claims are supported	Four to ten weeks, often longer
First decision	Editor, weighing the reports	Reject, major revision, minor revision	With the reports
Revision and response	You	What changes, what you defend	Weeks to months — usually the longest stage you control
Re-review and acceptance	Editor, sometimes the same reviewers	Accept, further revision, reject	Weeks

The time from submission to first decision is reported by many journals on their own pages and is worth looking up for the venue you have in mind: the answer is typically one to three months, and knowing it is how you plan around a thesis deadline.

Desk rejection is the editor declining without review, and it usually happens for one of three reasons: the work is out of the journal's scope, the advance is too incremental for that venue's threshold, or the manuscript is presented so poorly that the editor cannot see the claim. The first two are editorial judgements you can anticipate by reading the journal; the third is avoidable and is what chapter 63 is about.

A preprint changes three things: it timestamps the work publicly, it lets others read and cite it before review, and it exposes it to comment. It does not change whether the work is peer reviewed, how a hiring or funding panel weighs it in most systems, or your obligation to disclose it on submission. Most journals in this field accept preprinted manuscripts; a few do not, and that is worth checking before posting rather than after.

Reviewing a paper properly

The structure of a useful referee report, which is also the structure of how you should read any paper in your log:

- **Summary in your own words** — two or three sentences on what was done and claimed. If you cannot write it, you have not understood the paper, and saying so is itself a finding.
- **The claim**, stated as the authors would state it, so that everything after it is judged against that and not against the paper you would have written.
- **Do the data support it?** — the central question, and the one that makes the rest subordinate.
- **Controls** — part 7 chapter 22's list: the no-template, the empty vector, the loading control, the non-targeting guide, the parent strain assayed alongside.
- **Reproducibility** — could you repeat the method from the methods section, and are the data and code available (part 10 chapter 36)?
- **Limitations** — stated by the authors, or not.
- **Recommendation**, with major and minor points separated.

The discipline that makes a report useful rather than merely critical: **every major point must name what would fix it**. "The claim is unsupported" helps nobody; "the claim that flux was redirected needs either a carbon balance including by-products or a labelling experiment, since product up with uptake also up is consistent with higher throughput" tells the authors exactly what to do, and it is the standard this module has used since part 2 chapter 11.

Then the two questions you could not answer from the paper as written. Those are the most valuable sentences in a

report, because they are what the next reader will also not be able to answer.

Responding to reviewers

The response is a document with its own conventions: every point quoted, every reply saying **what changed and where** — page and line — or why you disagree, with evidence. Three habits make the difference. **Never ignore a comment**, including one you think is foolish; an unanswered point reads as an evasion. **Disagree in evidence, not in tone**: "we respectfully disagree; the control the reviewer asks for is shown in Figure S3, which we have now referenced in the main text" is a disagreement that wins. And **make the revision findable**: a marked-up manuscript plus a response that points to it costs you an hour and saves the editor an argument.

If a reviewer asks for an experiment you cannot afford, you have four honest moves and should choose explicitly rather than hoping: do a cheaper experiment that addresses the same doubt and say so; show that existing data already answer it; narrow the claim so the experiment is no longer needed — often the best option, and the one that improves the paper; or argue that it is beyond the scope, which works only if you also narrow the claim. What does not work is promising it as future work while keeping the strong claim.

Authorship and integrity

The widely used criteria — the ICMJE conditions — require **all** of: substantial contribution to conception, design, acquisition, analysis or interpretation; drafting or critically revising the intellectual content; approval of the final version; and agreement to be accountable for the work. Contribution alone is not enough, and neither is seniority. **First author** signals the person who did most of the work and usually wrote the draft; **corresponding author** signals who is answerable for the work and handles the submission, normally the supervisor. Order between the two is negotiated, and the time to negotiate it is at the start of a project, not at submission.

The three cases the week sets, with my answers and the reasoning:

Case	Decision	Why
A technician who ran every assay	Usually authorship; at minimum a substantial acknowledgement	Data acquisition is a listed contribution. The question is whether they also contributed intellectually and can approve and defend the work — if they designed or interpreted the assays, authorship is clear; if they executed a protocol exactly as specified and will not review the manuscript, a detailed acknowledgement is the honest answer. The failure mode to avoid is excluding them by default because of their job title.
A collaborator who supplied a strain	Normally acknowledgement, not authorship	Providing a reagent is not conception, analysis or interpretation. It becomes authorship if they also contributed to design or interpretation — for instance by advising on how the strain behaves in a way that shaped the experiments.
A supervisor who did not read the draft	Does not qualify as written, and the right response is to get them to read it	Approval of the final version and accountability are explicit criteria. In practice this is the most common gap between the standard and the norm, and the useful framing is not to police it but to insist on the reading — which is also what makes the paper better.

Image manipulation: adjustments applied uniformly to a whole image — brightness, contrast, colour balance — are legitimate if disclosed and if they do not obscure or create features. What is not legitimate: adjusting part of an image, splicing lanes without a visible divider and a note, removing a blemish, duplicating a region, or enhancing a band. Journals increasingly screen for this automatically, and the expectation is that you keep the unprocessed originals. The rule that covers every case: a reader must be able to reach your conclusion from the raw data, and any processing must be reversible in description.

The other integrity failures are statistical and were covered quantitatively earlier: **selective reporting** and **p-hacking** (part 2 chapter 17's multiplicity arithmetic, where 96 tests at $\alpha = 0.05$ give five false positives by construction), and **salami slicing**, dividing one result into several papers, which inflates a publication list and wastes readers' time.

Negative results matter for a reason this module has demonstrated repeatedly: part 8 chapter 29's honest base rate for pump engineering came from a screen that found nothing useful for two of seven compounds, and part 11 chapter 39's aerobic search found no coupled design at all — which was the result. A field that publishes only successes has a literature in which every intervention appears to work, so the next student repeats the failures privately. Reporting a negative result properly means stating what you would have been able to detect — part 2 chapter 17's power argument — which converts "it did not work" into a measurement.

Reading a lab through its papers

For each lab on your shortlist, count: papers in the last three years, where, and how often a student is first author. The pattern tells you things no website does.

Pattern	What it suggests
Many papers, students rarely first author	A large group where students contribute to others' projects; you may publish often and own little
Few papers, students always first author	Projects are owned and long; higher risk per student, more credit when it works
Consistent venue and topic	A defined programme; you will join a line of work rather than start one
Scattered topics and venues	Either a broad group or one following funding; ask which
Long gaps then a cluster	Projects that publish at the end — worth knowing before you are three years in

The question to ask a current student: *how long was it from starting your project to your first submission, and what delayed it?* The question to ask the supervisor: *what would a successful first year look like, and what would be publishable from it?* Both answers are concrete, and a supervisor who cannot answer the second has not thought about your project as a sequence of outputs.

Route	`submit → desk check → review → decision → revise → accept; one to three months to first decision`
Desk rejection	`scope, insufficient advance, or an unreadable claim`
Review report	`summary · claim · support · controls · reproducibility · limitations · recommendation`
Major point rule	`every major point names what would fix it`
Authorship	`ICMJE: contribution AND drafting or revising AND approval AND accountability — all four`
Image rule	`uniform, disclosed, reversible in description; keep the originals`

A reviewer asks for a ^{13}C labelling experiment to support your claim that a deletion redirected flux. You have no access to the instrumentation. What do you write?

First decide whether the claim needs it, because that decides everything else. The claim "flux was redirected" does need evidence beyond a titer, as part 11 chapter 43 argued — but a closed carbon balance including by-products and biomass, with substrate uptake measured, distinguishes redirection from higher throughput, and that you can do. So the response runs: we agree that a titer alone does not establish redirection; we have added a carbon balance (new Table 2) showing that total uptake was unchanged while the by-product fell by the amount the product rose, which is consistent with redirection and inconsistent with increased throughput; a labelling experiment would additionally resolve the split at the branch point, and we state that limitation explicitly in the discussion.

If even the carbon balance is unavailable, the honest move is to narrow the claim to what the data support — "titer increased by X with no change in substrate uptake" — and say in the discussion what would establish redirection. That is a better paper than one whose central claim rests on an experiment the reviewer correctly says is missing, and reviewers generally accept a narrowed claim where they will not accept a defended overreach.

Three things people get wrong

- ‘The reviewer is wrong, so I will ignore that point.’ An unanswered point reads as an evasion. Disagree with evidence, in writing.
- ‘They contributed, so they are an author.’ The standard requires contribution *and* drafting or revising *and* approval *and* accountability.
- ‘Negative results are unpublishable.’ They are publishable when you state what you could have detected, which turns a non-result into a measurement.

Self-check — answer on paper first

1. Your paper is desk-rejected in four days. What are the likely causes and what do you do?

Scope, insufficient advance for that venue, or a claim the editor could not find. Read the rejection letter for which — editors usually say, briefly — and then check the journal's recent contents for whether work like yours appears there at all. If it is scope, resubmit elsewhere unchanged. If it is the advance, either add the experiment that raises it or choose a venue whose threshold matches. If it is presentation, the abstract and the first paragraph are where to work, since those are what a desk decision is made on, and chapter 63's four-move abstract is the fix.

2. A collaborator supplied a plasmid and asks for authorship. How do you handle it?

Apply the criteria rather than the relationship: supplying a reagent is not by itself conception, design, analysis or interpretation, so the default is a specific acknowledgement. Then ask whether there is more — did they advise on the construct's behaviour, interpret a result, or will they review and stand behind the manuscript? If so, authorship is

justified. Handle it by raising the criteria early and in writing, with the supervisor involved, because the conversation is much easier before the data exist than at submission.

3. What does a lab's publication pattern tell you that its website does not?

Whether students own projects — visible in how often a student is first author rather than third — how long projects take to reach an output, from the gaps between a student joining and their first paper, and whether the group has a programme or follows funding, from the consistency of topic and venue. It also shows where the group's work is read, which affects who sees yours. None of this is on a website, and all of it predicts your own years there more reliably than a research-interests paragraph.

What to open a book for. The ICMJE recommendations on authorship, which are short and are the standard most journals point at; COPE's guidance and flowcharts, which show how real cases are handled; and the literature on peer review and on research integrity that the tracker names. Then read one journal's own guide for authors and reviewers in full — it is the most concrete source there is.

Chapter 63 · tracker week 63

Writing and talking about your work

A reader is looking for the argument and will stop looking if it is hard to find. The craft is not elegance; it is making the structure so obvious that the reader cannot lose it.

The shape of an argument

The pattern that organises almost every good paper is **context, content, conclusion**, applied recursively: to the paper, to each section, and to each paragraph. Context says why this matters and what is known; content says what was done and found; conclusion says what it means. And **one idea per paragraph**, with that idea in the first sentence — because a reader skimming first sentences should get the argument, and a reader who does not skim loses nothing.

The exercise that matters is this: write your capstone's **single central claim in one sentence**, then list the evidence under it. If the sentence needs an "and" joining two claims, you have two papers or one unfocused one. Then **mark every place where you asserted something the evidence does not reach** — and there will be several, because enthusiasm writes faster than evidence. In my own writing across this module the recurring form of that error was a confident number from a model whose assumptions I had not stated beside it, which is why every chapter here carries its parameters.

Two specific moves that make a metabolic-engineering argument legible, both of which the module has used throughout: state the **number that decides the question** early — the ceiling, the control coefficient, the k_La — because a reader who knows what the binding constraint is can follow everything after it; and **name the alternative you rejected**, which is what distinguishes a choice from a default, exactly as part 6 chapter 61 required for a host.

Figures that argue

A figure should carry **one message**, which should be the title or the first clause of the caption. The audit to run on your own figures:

- **Axes** with units and a readable range; log scales labelled as such, since a log axis flatters a small difference and a reader must be told.
- **Error bars defined** — standard deviation or standard error, and n, in the caption. Part 4 chapter 17's argument: they differ by √n and a caption that omits which is unreadable.
- **No decoration**: no three-dimensional bars, no gradients, no gridlines competing with the data, no colour carrying meaning that is not also carried by shape or position.
- **Every element earns its place**: if removing something changes no message, remove it.
- **A caption that stands alone**, including what was measured, n, what the error bars are, and the assumptions behind any computed line.

Rebuild the one figure that carries the capstone's argument to that standard. For a metabolic-engineering proposal that figure is usually the yield or flux analysis: the theoretical ceiling, the model prediction under a realistic constraint, and where the published state of the art sits — three numbers on one axis, which is an argument rather than a decoration.

Abstract and title

The four moves of an abstract, in order, with roughly a quarter of the words each: **problem** — what is wrong or unknown and why it matters; **approach** — what you did, specifically enough to be recognisable; **result** — the number, not the adjective; **consequence** — what follows.

Then cut it by a third. The words that go are the hedges, the throat-clearing first sentence, the restatement of the approach in the result, and every "novel" and "promising". The test the week sets is the right one: give it to someone unfamiliar and ask them what you are claiming. If their answer is not your central claim, the abstract has failed regardless of how well written it is.

Three candidate titles for a proposal on, say, a C4 diacid:

Type	Example	Judgement
Descriptive	"Growth-coupled production of a C4 diacid in *E. coli*: design, modelling and scale-up analysis"	Safe, searchable, says what is inside. Submit this.
Claim-making	"Oxygen transfer, not pathway flux, sets the ceiling for C4 diacid production"	Stronger if the evidence supports it — and in a proposal it usually does not yet, since the claim is what you propose to test

Too clever	"Breathing room: the hidden cost of aerobic diacid production"	Unsearchable and will read badly in a citation list

For a proposal, descriptive; for a paper whose central result is genuinely a reversal of the expected constraint, claim-making, because that is the finding. Never the third.

The ten-minute talk

Ten minutes is **one idea**. The structure that survives: thirty to sixty seconds of context — the minimum needed to make the question sensible, not a review; the question in one sentence on its own slide; the approach; the result, with the figure you rebuilt; and the limitation, **raised by you before anyone asks**. That last move is the one that changes how a room treats you: naming your own weakest point converts a cross-examination into a discussion, and it is the same argument as part 11 chapter 39's reporting of a search that found nothing.

Record it and watch it back. The two things to cut are everything you **skipped past** — a slide you waved at is a slide you do not need — and everything you **apologised for**: "sorry, this is a busy figure" means rebuild the figure, not warn the audience.

On questions you cannot answer, which is the week's last skill. The honest sequence: say you do not know; say what you would do to find out; and, if you have a partial answer, mark clearly where the evidence stops and the speculation begins — "what I know is X; what I would expect is Y, and the experiment that would settle it is Z." That answer is better received than a confident guess, and in a PhD interview it is close to the only answer that is.

Structure	`context · content · conclusion, recursively; one idea per paragraph, in the first sentence`
The test of a claim	`one sentence, no "and"; evidence listed under it`
Figure	`one message · units · defined error bars with n · nothing decorative · standalone caption`
Abstract	`problem · approach · result · consequence, then cut by a third`
Talk	`one idea; raise your own limitation; cut what you skipped or apologised for`
Unanswerable question	`I don't know · here is how I'd find out · here is where evidence stops`

You have ten minutes to present the capstone to a group that works on a different host. What goes in, and what gets cut?

In: the target and why it needs engineering rather than chemistry, in two sentences; the host choice with the deciding property named and the alternative rejected; the yield analysis figure — ceiling, constrained prediction, state of the art; the one experiment that would falsify the design; and the limitation you would raise yourself, which for most proposals is either a step with no good candidate enzyme or an oxygen-transfer ceiling.

Out: the pathway diagram with every intermediate labelled, which costs two minutes and conveys less than "four steps from acetyl-CoA, all with characterised enzymes except step three"; the literature survey; the methods you have not yet used; and every slide that exists because it was work. The hard cut is the last one — the modelling notebook took a week and gets one figure, because the audience is deciding whether the design is sound, not whether you can use COBRApy. Point at the repository for the rest, which is what artifacts are for.

Three things people get wrong

- ‘The reader will follow the logic.’ They will follow the first sentences of paragraphs. Put the idea there.
- ‘The figure shows the data.’ A figure should show one message; data with no message is a table.
- ‘I'll handle the weakness if someone asks.’ Raise it first. It converts an interrogation into a conversation.

Self-check — answer on paper first

1. Write the four moves of an abstract for a result you already have — the part 10 core model will do.

Problem: constraint-based models are used routinely to predict yields and deletions, but their assumptions are hidden inside software. Approach: a 40-reaction core model of *E. coli* central metabolism was built by hand and every standard analysis — flux balance, variability, deletions, parsimonious flux, shadow prices and quality control — implemented directly as linear programmes. Result: the model grows at 0.754 h^{-1} aerobically and 0.217 anaerobically, ten of 39 deletions are lethal, fourteen reactions have alternative optima at exactly maximal growth, and the implementation exposed three modelling errors that a package would have hidden. Consequence: writing the programme yourself is a better first exercise than using a package, because every later failure is a bound, a medium or an objective.

2. Your figure has six panels and reviewers keep asking what it shows. Diagnose it.

Six panels is at least two figures and possibly six: a figure carries one message, and a reader asking what it shows is telling you the message is not identifiable. Split it by message, promote the panel that carries the central claim to its own figure, and move the rest to supplementary material unless each is load-bearing. Then write each caption as a standalone sentence beginning with the message — if you cannot, that panel has no message and should go.

3. In a talk, someone asks a question that reveals a flaw you had not considered. What do you say?

Acknowledge it directly — "that is a real problem and I had not considered it" — then think aloud about what it would take to find out: which measurement would distinguish the possibilities, and what you would expect under each. Do not defend, do not improvise a mechanism, and do not promise it is fine. The reason this is the right answer is that the room is evaluating how you handle being wrong, which in research is the more frequent condition, and a candidate who can say "I don't know, here is how I'd find out" is demonstrating the thing the job actually requires.

What to open a book for. The ten simple rules for structuring papers, which is short and is the standard reference for the context-content-conclusion pattern; a figure-design guide; and then your own figures from across this curriculum, audited against the list above — that audit is the content of the week, not the reading.

Chapter 64 · tracker week 64

Planning a multi-year project: the critical path and what you cannot control

A plan built from most-likely durations is usually missed, and by a predictable amount. One simulation shows why, and it is the week's whole argument.

From an aim to a task list

The work breakdown: every task gets a **duration**, its **predecessors**, and a **deliverable**. The deliverable matters because it is what makes a task finishable — "characterise the strain" is not a task, "a table of titer, yield and growth rate for three strains in triplicate" is.

Here is the first year of a capstone-style project as ten tasks, with optimistic, most-likely and pessimistic durations in weeks. Two are **waiting rather than working**, which is the distinction the week insists on:

Task	o / m / p (weeks)	Predecessors	Kind
A — literature review and design	2 / 3 / 5	—	work
B — order gene synthesis	3 / 4 / 8	A	**waiting**
C — build vectors	1 / 2 / 4	B	work
D — host toolkit (parallel)	3 / 5 / 10	A	work
E — transform and verify	1 / 2 / 5	C, D	work
F — shake-flask screen	2 / 3 / 6	E	work
G — analytical method	2 / 3 / 7	A	work
H — quantify and iterate	3 / 4 / 8	F, G	work
I — fermenter slot	2 / 4 / 12	H	**waiting**
J — fed-batch and write-up	3 / 4 / 7	I	work

The critical path, and the float nobody uses

With most-likely durations the project takes **26 weeks** and the critical path is A → B → C → E → F → H → I → J. The two off-path tasks have float: the host toolkit has **1 week** and the analytical method has **8 weeks**. Shortening the analytical method by a month changes the end date by nothing at all, which is the first lesson and the one most people violate by optimising whatever is in front of them.

But float computed from single durations is misleading, because durations are uncertain. Sampling each task from a triangular distribution over its three estimates and recomputing the critical path 20,000 times gives the **criticality index** — the fraction of runs in which a task lies on the critical path:

Task	Criticality	Reading
A, E, F, H, I, J	100%	Always critical — attention here always matters
B order synthesis, C build vectors	76%	Usually critical
D host toolkit	**24%**	The near-critical branch that quietly becomes the problem
G analytical method	0%	Never critical, despite 8 weeks of float being only nominal

The host toolkit has one week of nominal float and is on the critical path in a quarter of simulated runs, because its pessimistic duration is ten weeks. That is the "second near-critical branch" the week warns about: a task you have classified as non-critical and therefore stopped watching.

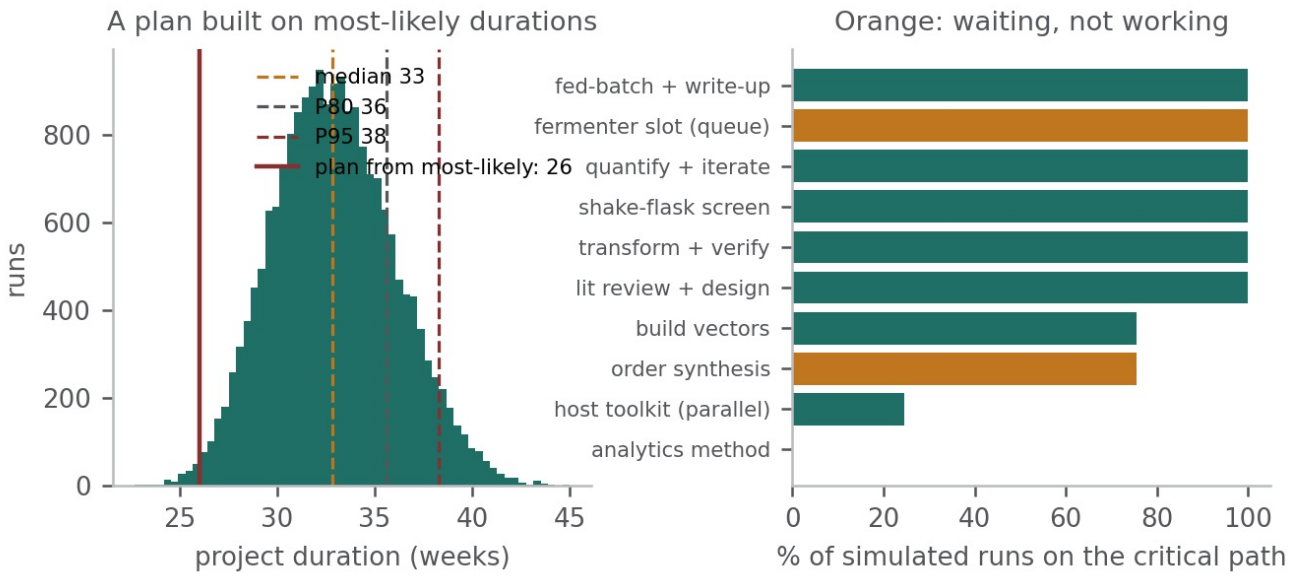


Figure 64.1 Left: simulated project duration over 20,000 runs with triangular task durations, against the 26 weeks a most-likely plan predicts. Right: criticality index per task, with the two waiting tasks in orange. Computed for this module from the task list above; the durations are illustrative and the triangular distribution is the assumption.

Why plans slip

The distribution is the week's central result:

Plan from most-likely durations	26.0 weeks
Simulated median	32.8 weeks
80th percentile	35.6 weeks
95th percentile	38.3 weeks
Probability of finishing within the planned 26 weeks	**1%**

A plan built by adding most-likely durations along the critical path has a one-in-a-hundred chance of being met, and the median overrun is 26%. Two mechanisms produce this and both are structural rather than behavioural. **Task durations are right-skewed**: a task can finish a little early and can be delayed a great deal, so its mean exceeds its mode. And **paths merge**: task E waits for both C and D, so it starts when the *later* of the two arrives, and the maximum of two uncertain quantities is larger than either's typical value. Every merge point in a network adds a little delay that no individual estimate contains.

So: **the date you put in a proposal and the date you privately expect should differ**, and you should know which is which. Put a date near the 80th percentile in the proposal — 36 weeks here — and plan your own work against the median. A proposal promising the 26-week date is not optimistic, it is innumerate, and a supervisor who has run projects will read it as such.

Where to spend attention: waiting against working

The week asks what halving a waiting time is worth against working faster. Simulated on the same task list:

Intervention	Median	Saving	80th percentile
Baseline	32.9 weeks	—	35.6
Halve the synthesis wait	31.5	1.4 weeks	34.2
Halve the fermenter queue	29.9	**3.0 weeks**	32.1
Halve both waits	28.5	4.3 weeks	30.7
Work 30% faster on every bench task	26.0	**6.8 weeks**	28.5

The honest result, which is not quite the one the week implies: in *this* task list, working 30% faster on everything saves more than halving both waits, because the waits are only 8 of the 26 nominal weeks. But the two interventions are not comparable in difficulty. Halving a fermenter queue means booking earlier or finding a second vessel — a phone call and

some planning. Working 30% faster on every bench task for nine months is not a decision, it is a different person. **Judge interventions by effort as well as by weeks**, and on that basis the queue is the best value on the table: three weeks for an email.

The general rule that survives: compute the criticality index, then look first at the critical tasks you do not control, because those are the ones where a small amount of foresight — ordering earlier, booking the slot before you need it, starting the approval paperwork in week one — buys weeks for nothing.

Buffers

Two ways to protect the same 5.2 weeks of padding on a 26-week plan: add 20% to every task, or hold one buffer of 5.2 weeks at the end. Simulated with the padded tasks subject to Parkinson's law — a task with a padded estimate takes at least that long, because work expands to fill the time available:

Approach	Median duration	Probability of meeting the 31.2-week committed date
Pad every task by 20%, pads consumed	35.2 weeks	**0%**
One buffer of 5.2 weeks at the end	32.8 weeks	**30%**

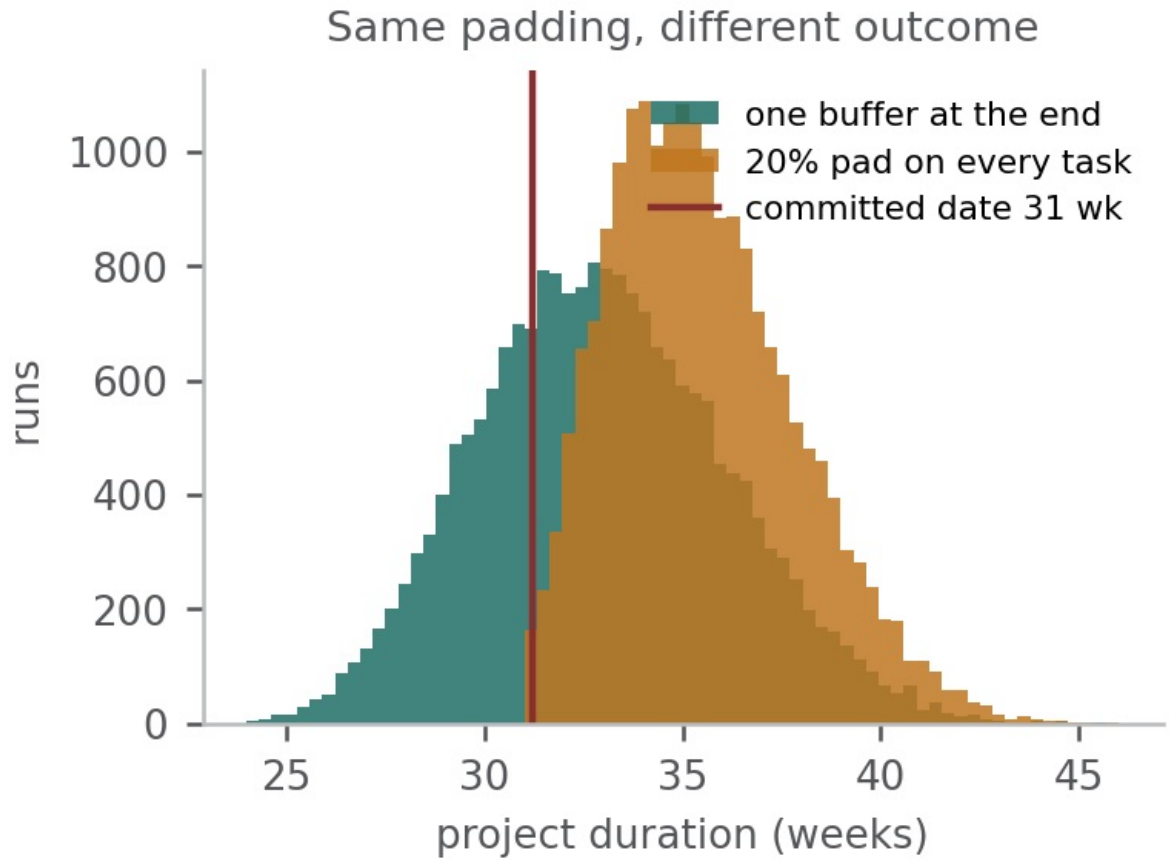


Figure 64.2 The same total padding, applied per task and consumed, against held as a single buffer at the end. Computed for this module; Parkinson's law is modelled as each padded task taking at least its padded estimate, which is an assumption about behaviour rather than a measurement.

Distributed padding is strictly worse, for two reasons. The pads are **consumed** rather than saved, because an early finish on a padded task is not passed on; and they are **invisible**, so nobody can see how much protection remains. A single buffer is visible, is consumed only by real overruns, and can be reported: "we are twelve weeks in, two weeks of the five-week buffer are gone" is a sentence a supervisor can act on.

And what to tell a supervisor three months into a six-month slip: the slip, the cause, the revised date with its uncertainty, and the decision you are asking for. Not an apology and not a promise to catch up — the arithmetic above says catching up does not happen. The useful version is: *the strain work is eight weeks behind because the toolkit took ten weeks rather than five; the revised 80th-percentile date is X; the options are to drop the second host, to run the fermentation with the existing strain, or to extend — I recommend the second, and here is why.*

Money, and what a student controls

Price one month of the planned work. The categories, with the pattern rather than numbers I would have to invent: **large and obvious** — gene synthesis, which scales with length and is the single biggest line for a pathway project; sequencing; enzymes and kits. **Hidden** — plasticware, media components, antibiotics, and the shipping and import charges on everything, which in a non-US lab can be a quarter of a reagent's cost and weeks of its lead time. **Shared facility charges** — analytical time per sample, fermenter day rates, sequencing per lane — which are billed to the project but queued by the facility, and therefore appear in both the budget and the schedule.

Look up current prices for your own institution rather than taking any figure from me: synthesis and sequencing prices in particular have moved by orders of magnitude over a decade, and a stale number in a proposal is a visible error. What a student actually controls is a small discretionary fraction; the rest is committed by the grant, which is why the next paragraph matters.

To see what funds a lab, read the **acknowledgements** of its recent papers: they name the programme and often the grant number. Then read what that programme is for — its call text states objectives in plain language — and write one sentence connecting your capstone to them. That sentence is worth disproportionate attention in an application, because it tells a supervisor that your project can be funded from money they already have.

Risk, and deciding in advance when to stop

Risk	Likelihood	Cost if it happens	Alternative
A pathway step has no active enzyme in the host	Moderate to high — part 11 chapter 40's p_n	Months; the central claim fails	Screen homologues; switch to the second route identified in the proposal
Product is toxic below the needed titer	Moderate	Caps the titer; TEA may fail	Two-phase removal (part 3 chapter 45); tolerance evolution
Oxygen transfer limits at the target density	Known in advance from chapter 44's arithmetic	Lower productivity, by-products	Lower feed rate, enrichment, or a lower-density higher-q_P design
Host toolkit takes longer than planned	24% critical in the simulation	Weeks on the critical path	Prototype in *E. coli* and transfer (part 7 chapter 25)
Strain instability at scale	Moderate	Invalidates the fermentation result	Integrate rather than carry a plasmid; sequence after the run (part 9 chapter 32)

Three go/no-go points with the triggering result, which is the discipline that stops two years disappearing into one idea: **month 4** — if no construct expresses a detectable product, stop and test whether the enzymes are active in lysate before building more strains; **month 8** — if the best strain is below a stated fraction of the modelled yield with no identified bottleneck, switch to the alternative route; **month 12** — if the techno-economic sketch at the achieved yield and titer cannot clear the market price under favourable assumptions, change the target rather than the strain.

And the sentence about risk for the proposal, which is the week's last point: *the highest risk is that step three has no characterised enzyme active on our substrate; we will test this in month two by expressing three homologues and assaying the lysate, and if none is active we will pursue the alternative route via [X], which is two steps longer but fully characterised.* Naming a risk strengthens a proposal because it demonstrates that you know where it might fail and have thought about what to do — which is indistinguishable, to a reader, from having run projects before.

Critical path	`the chain that sets the end date; float elsewhere is free time you cannot spend on it`
Criticality index	`fraction of simulated runs on the critical path; 24% for a task with 1 week of nominal float`
Why plans slip	`right-skewed durations and merge points: median 32.8 against a 26-week plan, P(met) = 1%`
Proposal date	`the 80th percentile; plan privately against the median`
Buffers	`one at the end, visible and managed, not 20% on every task`
Risk table	`what could stop it · likelihood · cost · alternative · and the result that triggers a stop`

Your supervisor asks for a one-year plan with a date. What do you give them?

Two dates and the reason they differ. The committed date at roughly the 80th percentile of a simulated schedule — in my example 36 weeks against a 26-week most-likely plan — with a single visible buffer that accounts for the difference; and the median as the internal working target. Alongside them: the critical path, the criticality index so they can see which tasks matter, the two or three tasks that are waiting rather than working with their lead times, and the go/no-go points with the results that trigger them.

What I would not give them is a Gantt chart of most-likely durations, because its date is met 1% of the time and everyone involved will spend the year explaining why it is late. The conversation to have instead is which of the three or four interventions is worth buying: in the simulation, halving the fermenter queue was worth three weeks for an email, while working 30% faster everywhere was worth seven weeks and is not available. Framing the plan as a set of purchasable improvements rather than a promise is the thing your engineering background lets you do that most applicants cannot.

Three things people get wrong

- ‘The plan says 26 weeks.’ It says the most-likely durations sum to 26; the simulation says the median is 33 and 26 is met 1% of the time.
- ‘That task has float, so it does not matter.’ One week of nominal float and a long tail made it critical in 24% of runs.
- ‘I padded every task to be safe.’ Padding is consumed and invisible; one buffer at the end gave 30% on-time against 0%.

Self-check — answer on paper first

1. Why does adding most-likely durations along the critical path underestimate the duration?

Two reasons. Task durations are right-skewed — a task can run a little early and a long way late — so each task's mean is greater than its most likely value, and the sum of means exceeds the sum of modes. And wherever paths merge, the successor waits for the *latest* predecessor, so the expected start is the expectation of a maximum, which is greater than the maximum of the expectations. Every merge point adds delay that no individual estimate contains, which is why the simulated median exceeded the deterministic plan by 26% here.

2. A task has four weeks of float. Under what circumstances should you still watch it?

When its distribution has a long right tail, which is exactly the host-toolkit case: one week of nominal float, a pessimistic duration twice the most likely, and a 24% criticality index. Float computed from single-point estimates tells you about the mode, not the tail, and a task that is non-critical in the plan and critical in a quarter of futures deserves attention. The quantitative test is the criticality index, and the practical sign is a task whose pessimistic estimate is more than about twice its most-likely one.

3. You are eight weeks behind at month six. What do you tell your supervisor?

The slip, its cause, a revised date with its uncertainty, and a decision to be made — with your recommendation. Not an apology, and not a plan to catch up, because the arithmetic says recovery does not happen: the remaining tasks have the same distributions they always had. The useful content is the options and their consequences — drop a scope item, run with what exists, or extend — each with what it costs, so that the person who has to decide can decide. And if a buffer was declared at the start, say how much of it is left, which turns the conversation from blame into inventory.

What to open a book for. No book: the tracker's literature searches on research project management and on Monte Carlo schedule risk analysis, plus the NumPy random-number documentation, are what this week needs. The simulation above is thirty lines, and building your own on your own task list is the content of the week.

The capstone proposal, section by section

Three to five pages that a principal investigator reads in ten minutes and either believes or does not. What makes it believable is not the science, which anyone can read up — it is that the numbers are yours and the risks are named.

What the proposal is for

It is an argument that a specific molecule should be made in a specific host by a specific route, that the numbers support it, and that you know where it will fail. Seven sections, roughly: motivation, host, pathway, the numbers, build-and-test, scale-up and economics, risks. The order matters because each section's conclusion is the next one's premise — the molecule sets the price it must clear, the price sets the titer and yield required, those set the host and the route, the route sets what must be built, and the economics decides whether any of it was worth it.

Section 1 — target and motivation, one page

Why this molecule: what it is used for, the market size with a source, and **the incumbent route**, because that is what you are competing against. Part 5 chapter 53's lesson is the governing one — a target whose incumbent can halve its price in response is a bad target however good the biology — so name the incumbent's cost structure, not just its existence: a petrochemical plant with amortised capital, an agricultural supply that responds to price, or an extraction from a scarce natural source that cannot scale.

Then the sustainability argument, **honestly, including where it is weak**. The honest version almost always includes: the feedstock is agricultural and therefore carries land and fertiliser burdens; the electricity mix decides the result (part 5 chapter 57's 17% advantage that a grid at 0.6 kg CO_2e/kWh erases); and the boundary you chose is the one that makes your number look as it does. A proposal that states this is more credible than one claiming a clean win, and a reviewer goes to the boundary first anyway.

Section 2 — host, with the deciding property named

Use your own host table from part 6 chapter 61, and write the choice in the format that chapter required: **the property that decides it, the alternative you rejected, and what you gave up**. "We will use *K. phaffii* because the product is a secreted haem protein and this host secretes at grams per litre with a clean extracellular background and a food-enzyme regulatory history; we rejected *E. coli*, which would have given a faster design–build–test cycle and a far better toolkit but requires periplasmic targeting or refolding and carries endotoxin a food product cannot have; what we gave up is cycle time."

Then the **localisation subsection**, which part 6 chapter 59 argued decides more than it is given credit for: secreted, retained or stored. Secreted means no intracellular ceiling and a simpler downstream train. Retained means the cell-volume ceiling of part 1 chapter 1, recomputed at industrial density — 20 g/L at 50 gDCW/L, 40 at 100 — and therefore a high-cell-density process plus lysis and debris removal. Stored means an accumulation physiology and a nutrient-limitation trigger (part 5 chapter 56). Say which, and say what it forces on the host.

Section 3 — the pathway, every enzyme sourced

From a named central metabolite — one of the twelve of part 1 chapter 7 — to the product, step by step, with an enzyme named and cited for each from BRENDA, MetaCyc or KEGG, including the organism it came from and, where known, its activity on a substrate like yours. Then the sentence that makes this section honest: **mark the steps with no good candidate enzyme**. Part 11 chapter 40's arithmetic says a route of n steps each with probability p works with probability p^n, so five steps at p = 0.8 is 33%, and a step with no candidate at all is a discovery project rather than a low probability. Marking them is not a weakness in the proposal; it is the difference between a route and a wish.

Section 4 — the numbers, worked by hand and then in a model

This is the section that distinguishes your proposal from a biologist's, and it has three parts.

The hand calculation. Theoretical maximum mass and molar yield on glucose, with the carbon, NADH, NADPH and ATP balances closed *explicitly and separately* — part 2 chapter 19's order, and the point at which most back-of-envelope balances fail by lumping the two hydride carriers. Show the working: part 5's ethanol at 0.511 g/g with its redox closing exactly, isobutanol at 0.411 with its cofactor mismatch, PHB at 0.478 with a surplus of three reduced cofactors and therefore an oxygen duty, are the models for how this looks.

The model. Add the heterologous reactions to a genome-scale model — metabolites, stoichiometry, bounds, objective — and **confirm it reproduces the hand yield**. That confirmation is the point: if the model and the hand calculation disagree, one of them is wrong, and finding out which is the exercise. Part 10 chapter 36's warnings apply: set the medium through the exchange reactions, use a context manager for temporary changes, and check for the loops and alternative optima that make a single flux vector meaningless.

The constrained prediction. Then the step that almost no student proposal contains: constrain oxygen to a value your own k_La calculation justifies (part 11 chapter 44) and rerun. In my core model, going from unconstrained oxygen to the transfer a 100 h^{-1} vessel delivers at 10 gDW/L cost 58% of the growth rate and produced 12 mmol of reduced by-products. **Report the yield you would actually get in a vessel**, not the one the unconstrained model gives, and say which is which.

Push the notebook to the repository, commented so a PI can read it without running it, and put its headline figure — ceiling, constrained prediction, state of the art — in the proposal. That figure is the one to rebuild to chapter 63's standard, because it carries the argument.

Section 5 — build and test, with a reason per tool

Assembly method, editing tool, analytics, screening readout, and the number of design–build–test iterations you expect. The rule the week sets is the right one: **no tool without a reason, in one clause**. "MoClo-standard Golden Gate, because six parts will be swapped combinatorially (part 7 chapter 20); CRISPR–recombineering with Cas9 counterselection, because the host has no efficient end joining and the edits are deletions (chapter 23); CRISPRi for the two essential competing genes, because a knockout is not recoverable (chapter 24); HPLC with a weighted low-range calibration for titer, because the first strains will be near the quantification limit (part 4 chapter 50)."

Then the **transport strategy**, which part 3 chapter 45 argued belongs on the first page rather than the last: if the product is secreted, which efflux pump you would mine, express or evolve, and the base rate you expect from the published screens; if it is retained, the lysis and extraction train and what it costs. And state **what result would make you abandon the design** — a sentence most proposals omit, and the one that makes a reader trust the rest.

Section 6 — scale-up, downstream and the economics

Fermentation mode with a physiological justification rather than a conventional one (part 11 chapter 44): exponential fed-batch below the overflow threshold if the host has one, two-phase with nitrogen limitation if the product is stored, gas-phase considerations if it is a knallgas organism. The scale-up risks, which you can now name quantitatively: mixing time growing as $d^{2/3}$, the k_La wall at 20 to 30 gDW/L, and the extra generations in the seed train with their instability risk (part 9 chapter 32's plasmid-retention recursion).

The downstream train as a list of unit operations with the property each exploits, and the water: 1,000 divided by the titer in litres of broth per kilogram of product, which at 20 g/L is fifty kilograms of water to move per kilogram made. Then the back-of-envelope from part 4 chapter 49: feedstock cost as substrate price over yield, capital and fixed cost scaling as 1/titer, and **the titer below which the process cannot work** — which in that model, for a $1.80/kg commodity, was around 100 g/L. State it as a target, because it converts the whole proposal into a number the strain work has to hit.

Section 7 — risks and alternatives, named specifically

Part 64's table, filled for this project: what could stop it, how likely, what it would cost, and what you would do instead, plus the go/no-go points with their triggering results. Specific means naming the step, the organism and the measurement — "step three may have no active enzyme" rather than "enzyme activity may be insufficient".

Reading it once, aloud

The last session's instruction is to read the whole draft aloud and mark the weakest section. Reading aloud finds three things that silent reading does not: sentences you cannot finish in one breath, which are too long; claims you hesitate before, which are the ones the evidence does not reach; and transitions that do not exist, where you have put two paragraphs next to each other and hoped. The weakest section in most of these proposals is section 5, because it is a list of tools and the reasons are the thing that is missing — which is why the one-clause rule exists.

Section 1	`molecule · market · incumbent's cost structure · honest sustainability, with its weak point`
Section 2	`deciding property · rejected alternative · what you gave up · localisation`
Section 3	`central metabolite → product, every enzyme cited, unknown steps marked`

Section 4	`hand yield with C, NADH, NADPH, ATP closed separately → model reproduces it → oxygen-constrained rerun`
Section 5	`one clause of justification per tool; transport strategy; the abandonment criterion`
Section 6	`feeding from physiology · mixing and k_L a limits · water per kg · the titer below which it fails`
Section 7	`risks with likelihood, cost and alternative; go/no-go triggers`

A PI reads your proposal and says "this is a nice review of the field, but where is your contribution?" What went wrong?

Almost certainly sections 1 and 3 swallowed the document and section 4 is thin. The failure mode is writing what is known — the pathway, the host's properties, the published titers — at the length it deserves in a thesis introduction rather than in a four-page proposal, so that the reader finds a competent summary and no argument. The contribution in a proposal of this kind is not the pathway, which is assembled from the literature; it is **the analysis**: the closed balance done by hand, the model that reproduces it, the oxygen-constrained yield, the titer the economics demands, and the named risks with their triggers. Those are the parts nobody else's proposal has.

The fix is mechanical. Cut sections 1 and 3 to one page combined, with the enzyme table as an appendix. Expand section 4 until the three numbers — ceiling, constrained prediction, required titer — are each derived in the text and visible in one figure. And add the two sentences that are almost always missing: what result would make you abandon the design, and which assumption, if wrong, changes the conclusion. A reader who can see those two sentences knows they are reading an engineer's proposal rather than a summary.

Three things people get wrong

- ‘The pathway is the proposal.’ The pathway is the easy part and is mostly literature. The analysis is the contribution.
- ‘Quote the model's yield.’ Quote the oxygen-constrained yield and say what constrained it; the unconstrained number is the ceiling, not a prediction.
- ‘Naming a risk weakens it.’ A named risk with an alternative reads as experience; an unnamed one reads as inexperience the moment a reviewer thinks of it.

Self-check — answer on paper first

1. Your model gives a higher yield than your hand calculation. What do you do?

Find the discrepancy before writing anything, because one of the two is wrong and it matters which. The usual causes, in order: the model has a route your hand balance did not consider — often a transhydrogenase or an alternative cofactor specificity that relieves the redox constraint you imposed; the model's biomass reaction is not being used, so carbon that should go to cells is going to product; a loop or an alternative optimum is inflating the exchange flux (part 10 chapter 36); or the hand calculation lumped NADH and NADPH, which is the commonest arithmetic error and the one part 2 chapter 19 warns about. Reconciling them is the exercise, and the reconciliation is worth a paragraph in the proposal because it demonstrates you did both.

2. What makes a host justification defensible rather than a default?

Naming the property that decides it, the alternative you rejected, and what you gave up by rejecting it. "*E. coli* because it is standard" is a default; "*E. coli* because the design needs twenty DBTL iterations and it has the shortest cycle in biology, accepting that the product is retained and will need lysis and that endotoxin removal will be required if this ever goes to a clinical product" is a choice. The test is whether the sentence would still be true if you had chosen differently — if it would, it is not a justification.

3. Which section do you expect to be weakest, and how do you tell?

Section 5, the build-and-test plan, because it is where a list of tools is easiest to write and hardest to justify. Tell by applying the one-clause rule mechanically: go through every tool named and check that a reason follows it that refers to *this* project — the number of variants, the essentiality of the gene, the expected concentration range. Any tool whose clause would be identical in someone else's proposal is a default. Reading aloud finds them, because you will hesitate before the ones you cannot justify.

What to open a book for. Your own paper log, and two recent papers from each target lab — the proposal has to touch what they are actually doing. Everything else you need is in parts 1 to 11 of this module, and the cross-references in the section list above are the route back to each calculation.

Chapter 66 · tracker week 66

Applications: labs, statement of purpose, interviews

Evidence over enthusiasm. Every sentence that would survive unchanged in someone else's application is a sentence doing no work.

Finding the labs

Three sources, in this order. The **editorial board** of *Metabolic Engineering* and of one or two adjacent journals, which is a curated list of the field's active groups. The **last year of titles** in two journals, read as titles only, which takes an hour and shows you what the field is currently doing rather than what it did when the textbooks were written. And the **acknowledgements and affiliations** of the papers already in your log, which is how you find the groups whose work you have been reading without noticing whose it was.

Shortlist eight to ten groups with, for each, one recent paper and one sentence on why it interests you. Then cut to three or four you would actually join — and the cut is the work, because a list of ten is a list of names and a list of three is a decision. What should decide the cut: whether the group's current direction touches your proposal, whether students lead papers there (chapter 62's pattern analysis), whether the techniques you have been learning are the ones they use, and whether you would be glad to spend four years on their questions rather than your own, because that is what happens.

Reading their work properly

For each shortlisted lab, read one recent paper closely and log it. Then two specific outputs. Write **how your proposal touches that lab's current direction** — not the field's, the lab's, in one sentence naming the paper. And note **the one key figure** of each paper, because that is what you will be asked about and because identifying it forces you to decide what the paper's claim was.

The test, from part 6 chapter 61's checkpoint: if you cannot name each lab's recent paper and its key figure, that lab is not on your shortlist yet — it is on a list of names.

The statement of purpose

Five parts, and the discipline is that each must point at something that exists.

Part	What it does	The failure mode
The problem	A concrete problem you want to work on, specific enough to be arguable	"Sustainability" — a topic, not a problem
Why this field now	Why metabolic engineering is the way to solve it, and why the moment is right	A paragraph about the field's importance that any applicant could write
Evidence of preparation	What you have actually done, **named**: the notebooks, the reproduced design, the flux analysis, the proposal	Listing coursework, or claiming interest
Lab fit	Which group, which paper, which direction yours touches	Praise for the group's reputation
Goals	What you would want to be able to do in four years	A career ambition rather than a capability

Then the cut that matters: **remove every sentence that would survive in someone else's application**. "I am fascinated by the potential of synthetic biology to address climate change" survives in ten thousand applications and therefore conveys nothing. "I rebuilt a 40-reaction core model of central metabolism as a linear programme to understand where flux balance analysis's assumptions sit, and found three of my own modelling errors doing it" survives in one.

The bridge from chemical engineering, which is this curriculum's whole premise, should be made in the language of the thing you bring rather than the thing you lack: stoichiometric and energy balances closed explicitly, kinetics and rate laws, reactor design and residence time, transport and the k_La arithmetic, separations, and techno-economic analysis. Part 11 chapter 44's oxygen-bound calculation and part 4 chapter 49's minimum selling price are the two artifacts that demonstrate it best, because almost nobody entering from biology has done either.

Polish and send

Revise the proposal's weakest section — chapter 65 says which it usually is — and tighten the whole to three to five pages. Send it to one person for comment, which is worth more than three rounds of self-editing. Then two short emails to principal investigators, and the instruction is exact: **a real question, not an introduction**.

What distinguishes them: a real question refers to a specific result in a specific paper and asks something that could not be answered by reading it again. "I was struck that your 2026 strain reached X with no change in uptake rate — did the carbon balance close, or is some of the gain recovery of a by-product you had previously lost?" is a question a PI can answer in two lines and will remember. "I am very interested in your work on Y and would like to discuss opportunities" is not, and answering it costs them more effort than it conveys.

The interview drill

Three questions you should be able to answer cold, and what a good answer contains.

"Why metabolic engineering, and why you?" — sixty seconds, containing a concrete problem, the reason metabolic engineering is the way to solve it, and at least one artifact by name. The test, again, is whether it would survive unchanged in someone else's mouth.

"Walk me through a paper you have read." — pick one from the log and give it in the structure of chapter 62's referee report: what was claimed, whether the data support it, what the key figure shows, and the two questions you could not answer from the paper. The last part is what distinguishes a reader from a summariser.

"What would you do differently?" — about your own work, and the honest answer is the one to give. Across this curriculum mine would be: build the model before reading more about the pathway, because the balance tells you what matters; and measure the transport capacity before designing the knockouts, because part 11 chapter 39's export cap made a coupled strain infeasible outright. An answer that names a specific technical decision and what it cost is worth more than a reflection on time management.

And the question you cannot answer, which is chapter 63's skill and will come up: say you do not know, say how you would find out, and mark where the evidence stops.

The final checkpoint

Can you say in sixty seconds why metabolic engineering, and why you?

Your own answer, but it must contain a concrete problem, the reason this field is the way to solve it, and an artifact by name. If it would survive unchanged in someone else's application, it is not yet specific enough.

Can you name three PIs, their recent papers and how your proposal touches each?

If you cannot name each lab's recent paper and its key figure, that lab is on a list of names rather than a shortlist.

Is your repository readable by a stranger in five minutes?

Open it as a stranger would — README first. If it does not say what the project is, what to run and what the result was, fix that before anything else. Then check that it runs from a fresh clone, which is the only real test (part 10 chapter 36).

Does your statement of purpose cite an artifact rather than an ambition?

The test is whether the document points at something that exists — a notebook, a reproduced design, a proposal, a flux analysis — rather than at an intention.

Then the last instruction in the tracker, which is the one I would keep: reread your week 1 notebook page, and write what you will do in the next six months. The point of rereading it is to see how much of what looked difficult in week 1 is now routine, which is the only reliable evidence that the sixty-six weeks worked — and the point of writing the next six months down is that the curriculum ends and the work does not.

Finding labs	`editorial boards · a year of titles · the acknowledgements in your own log`
Shortlist test	`name the paper and its key figure, or it is a list of names`
Statement	`problem · why now · evidence (named artifacts) · lab fit · goals`
The cut	`delete every sentence that would survive in someone else's application`
Emails	`a real question about a specific result, answerable in two lines`
Interview	`why you (60 s, with an artifact) · a paper as a referee report · what you would do differently`

A PI replies to your email with "interesting — what would you actually do in the first six months?" What do you send?

The chapter 64 answer, compressed to a paragraph and a table. A task list for the first six months with durations, the two or three tasks that are waiting rather than working with their lead times, the critical path, and a committed date at the 80th percentile with the median as the internal target. Then the first go/no-go point: the result at month four that would make you stop and test whether the enzymes are active rather than build more strains.

What makes this the right reply is that it answers a question most applicants cannot. A biologist's first-six-months plan is usually a list of experiments in sequence; yours is a schedule with uncertainty, float and a decision point, which is the thing your previous career actually taught you and the thing the proposal's reader cannot get from another candidate. Keep it to half a page, attach the proposal, and offer the repository — and then the conversation is about the project rather than about you, which is where you want it.

Three things people get wrong

- ‘I will explain my enthusiasm for the field.’ Every applicant does. Point at something that exists.
- ‘I will email to introduce myself.’ Introductions cost the reader effort and convey nothing. Ask a real question about a specific result.
- ‘I will avoid mentioning weaknesses in interview.’ The strongest answer to "what would you do differently" is a specific technical decision and what it cost.

What to open a book for. The *Metabolic Engineering* editorial board and recent issues of *ACS Synthetic Biology*, read as lists of names and titles rather than as papers. Everything else this week is your own output: the log, the repository, the proposal and the statement.

Closing notes to part 12 — and to the module

The five weeks in one page

Week 62. The route from submission to acceptance and who decides at each stage; desk rejection's three causes; a referee report whose every major point names what would fix it; the four authorship conditions, all of which must hold; and reading a lab through its publication pattern, which tells you whether students own projects.

Week 63. Context, content, conclusion, recursively, with one idea per paragraph in the first sentence. One message per figure, error bars defined with n. The four-move abstract, then cut by a third. A ten-minute talk is one idea with your own limitation raised first, and the honest answer to an unanswerable question has three parts.

Week 64. The week with the arithmetic. A ten-task first year with a most-likely duration of 26 weeks had a simulated median of 32.8, an 80th percentile of 35.6, and a **1% chance** of meeting the planned date — because durations are right-skewed and paths merge. A task with one week of nominal float was critical in 24% of runs. Halving the fermenter queue was worth three weeks for an email. And the same padding held as one buffer gave 30% on-time delivery against 0% when spread across tasks.

Week 65. Seven sections, each one's conclusion the next one's premise. The contribution is section 4: the hand balance with the cofactors separated, the model that reproduces it, and the oxygen-constrained yield you would actually get in a vessel.

Week 66. Evidence over enthusiasm, artifacts by name, and a real question rather than an introduction.

On the numbers in this part

The schedule simulation is the only computed content here and it was run: ten tasks with triangular durations over 20,000 iterations, with the critical path and criticality index recomputed in each. The task list and its three-point estimates are illustrative — they are a plausible first year, not a measurement — and the Parkinson's-law model in the buffer comparison is an assumption about behaviour rather than something I measured. The qualitative results are robust to the estimates: a plan summed from modal durations is met rarely, merge points add delay no single estimate contains, and distributed padding is consumed. The specific weeks are not.

Costs in week 64 are deliberately described as patterns rather than figures, because synthesis and sequencing prices move fast enough that a number from me would be wrong by the time you read it.

The module, finished

Twelve parts, 66 weeks. Part 1 covered the biology foundations as a spine beside the textbooks; part 2 the modelling and control framework at twice that density; parts 3 and 4 in situ removal, immobilisation, economics, validation, omics and host defences; part 5 the five product classes with a constraint sheet each; part 6 the hosts assessed against those sheets; part 7 assembly, assays and genome engineering; part 8 parts, circuits, libraries and protein design; part 9 integration, sequencing, RNA-seq and the biofoundry; part 10 constraint-based modelling built from first principles; part 11 strain design, machine learning, labelling and the bioreactor; and this part, the profession.

What held it together, if anything did, was a single habit: **compute the number that decides the question, and state the assumption it rests on beside it**. The capture factor and the displaced working volume; the summation theorem and the ceiling it puts on a single enzyme; the 24 usable four-base overhangs; the proteome cost of a slow enzyme; the oxidative branch's half-carbon per NADPH; 0.67 against −0.44 for the same dataset under two splits; a k_La of 1,560 against 500 achievable; and a 1% chance of meeting a plan built from most-likely durations. None of those is hard to compute and each one changes a decision.

Three cautions to carry out of it. The arithmetic here is mine and was run in a container without network access, so where a number came from memory rather than a calculation I marked it, and those are the ones to verify — eQuilibrator for free energies, BioNumbers for physiology, the primary paper for anything attributed. The models are small and their parameters are stated; they are teaching instruments, not descriptions of *E. coli*. And the errors are mine: three in the core model alone, each described where it happened because finding them is the skill the module was trying to teach.

What is left is the part no document can do. Go and run the sessions.